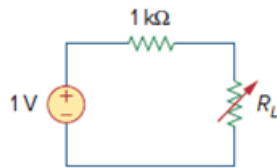


Problem

Find the maximum power transferred to R_L if the $1\text{-k}\Omega$ resistor in Fig. 4.55 is replaced by a $2\text{-k}\Omega$ resistor.

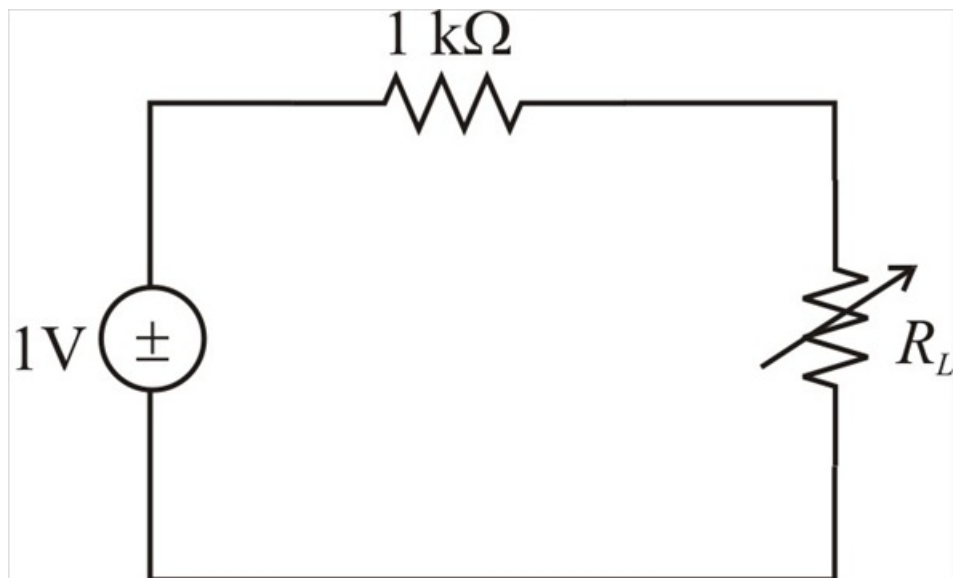
Figure. 4.55:



Step-by-step solution

Step 1 of 8

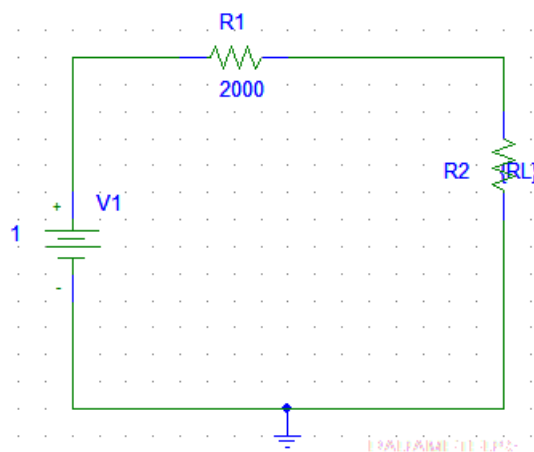
Considering the circuit



Step 2 of 8

We need to perform a DC Sweep on R_L to determine when the power across it is maximum.

We first draw the circuit using Schematics as shown in the following fig



Step 3 of 8

Once the circuit is drawn, we take the following three steps to further prepare the circuit for a DC Sweep.

The first step involves define the value of R_L as a parameter, since we want to vary it. To do this:

1. **DCLICKL** the value 1k of R_2 (representing R_L) to open up the *Set Attribute Value* dialog box
2. Replace 1k with $\{ R_L \}$ and click **OK** to accept the change.

Note that curly brackets are necessary.

Step 4 of 8

The second step is to define parameters. To achieve this

1. press **Ctrl+G**
2. Type **PARAM** in the *partName* box and click **OK**
3. **DRAG** the box to any position neat the circuit.
4. **CLICKL** to end placement mode.
5. **DCLICKL** to open up the *partName: PARAM* dialog box.
6. **CLICKL** on *NAME1=* and enter R_L (with no curly brackets) in the *Value* box, and **CLICKL Save Attr** to accept change.
7. **CLICKL** on *VALUE1=* and enter 2k in the *Value* box, and **CLICKL Save Attr** to accept change.
8. Click **OK**.

Step 5 of 8

The value 2k in item 7 is necessary for a bias point calculation; it cannot be left blank.

The third step is to set up the DC sweep to sweep the parameter.

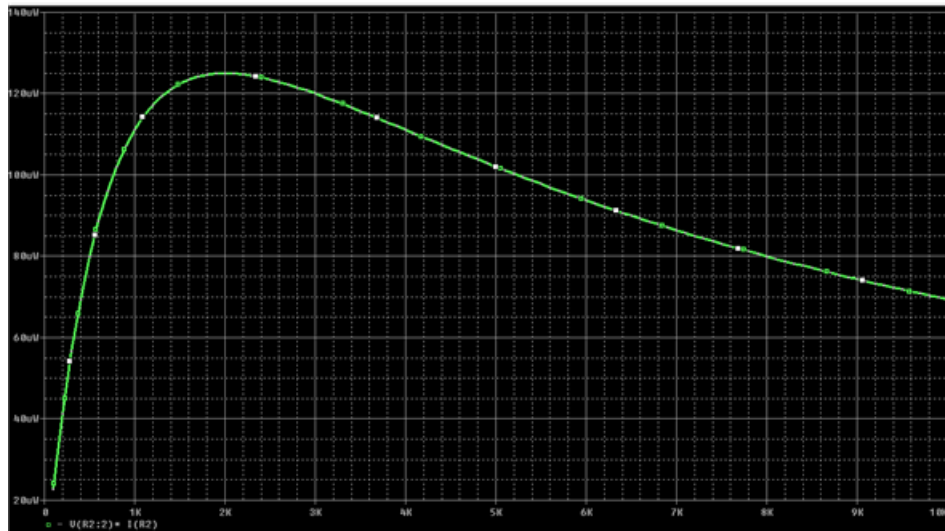
To do this:

1. Select **Analysis/Setup** to bring up the DC sweep dialog box.
2. For the *Sweep Type*, select Linear
3. For the *sweep Var. Type*, select Global Parameter.
4. Under the *Name* box, enter R_L
5. in the *Start Value* box, enter 100
6. In the *End value* box enter 10k
7. In the *increment* box, enter 100
8. Click **OK** and **Close** to accept the parameters

Step 6 of 8

After taking these steps and saving the circuit, se are ready to simulate. Select **Analysis/Simulate**. If there are no errors, we select **Add Trace** in the *PSpice A/D* window and type $-v(R2:2)*I(R2)$ in the *Trace Command* box. [The negative sign is needed since $I(R2)$ us negative]. This gives the plot of the power delivered to R_L as R_L varies from 100 to 5k.

Step 7 of 8



Step 8 of 8

It is evident from the plot that the maximum power is $125\mu\text{W}$. Notice that the maximum occurs when $R_L = 2\text{k}\Omega$, as expected analytically.